

4.19

Problem 1.

a.

$$\frac{}{\phi \vdash \text{True} : \text{Bool}} \quad \frac{}{\phi \vdash 0 : \text{Int}} \quad \frac{}{\phi \vdash \text{False} : \text{Bool}}$$

 $\phi \vdash \text{If True Then } 0 \text{ Else False} : \text{failed}$ since Int is not the same type as Bool

b.

$$\frac{}{x : \text{Bool} \vdash x : \text{Bool}} \quad \frac{}{x : \text{Bool} \vdash \text{True} : \text{Bool}}$$

$$\frac{}{x : \text{Bool} \vdash (x \text{ And True}) : \text{Bool}}$$

$$\frac{}{x : \text{Bool} \vdash x : \text{Bool}}$$

$$\frac{}{\phi \vdash \text{False} : \text{Bool}} \quad \frac{}{\phi \vdash (\text{Fun } x : \text{Bool} \rightarrow x \text{ And True}) : \text{Bool} \rightarrow \text{Bool}} \quad \frac{}{\phi \vdash (\text{Fun } x : \text{Bool} \rightarrow x) : \text{Bool} \rightarrow \text{Bool}}$$

$$\frac{}{\phi \vdash (\text{If False Then } (\text{Fun } x : \text{Bool} \rightarrow x \text{ And True}) \text{ Else } (\text{Fun } x : \text{Bool} \rightarrow x)) : \text{Bool} \rightarrow \text{Bool}} \quad \frac{}{\phi \vdash \text{True} : \text{Bool}}$$

$$\phi \vdash (\text{If False Then } (\text{Fun } x : \text{Bool} \rightarrow x \text{ And True}) \text{ Else } (\text{Fun } x : \text{Bool} \rightarrow x)) \text{ True} : \text{Bool}$$

$$\text{c. } \frac{}{x : (\text{Int} \rightarrow \text{Bool}) \vdash x : (\text{Int} \rightarrow \text{Bool})} \quad \frac{}{x : (\text{Int} \rightarrow \text{Bool}) \vdash \text{False} : \text{Bool}}$$

$$\frac{}{x : (\text{Int} \rightarrow \text{Bool}) \vdash x \text{ False} : \text{failed}}$$
 since Int is not the same type as Bool

$$\frac{}{x : \text{Int} \vdash x : \text{Int}} \quad \frac{}{x : \text{Int} \vdash \text{True} : \text{Bool}}$$

$$\frac{}{x : \text{Int} \vdash x \text{ And True} : \text{failed}}$$
 : needs both types to be Bool for And

$$\phi \vdash (\text{Fun } x : (\text{Int} \rightarrow \text{Bool}) \rightarrow x \text{ False}) : \text{failed}$$

$$\phi \vdash (\text{Fun } x : \text{Int} \rightarrow x \text{ And True}) : \text{failed}$$

$$\phi \vdash (\text{Fun } x : (\text{Int} \rightarrow \text{Bool}) \rightarrow x \text{ False}) (\text{Fun } x : \text{Int} \rightarrow x \text{ And True}) : \text{failed}$$

Problem 2.

a. neither

① suppose the left type is a subtype of the right :

we have a contradiction because the right record has the additional field w not in the left record

② suppose the right type is a subtype of the left :

we also have a contradiction because $\{ \} <: \{ z : \text{Bool} \}$ is not satisfied

($\{ z : \text{Bool} \}$ has the additional field z not in $\{ \}$), which is part of the definition ($T_2 <: T_1'$) for the right type being a subtype of the left

so the left type is neither a supertype or subtype of the right type.

b. supertype:

$$\frac{}{\vdash \{ x : \text{Int} \} <: \{ \}} \quad \frac{}{\vdash \{ x : \text{Int} \} <: \{ \}}$$

$$\vdash \{ \} \rightarrow \{ x : \text{Int} \} <: \{ x : \text{Int} \} \rightarrow \{ \}$$

(every record is a subtype of the empty record according to our subtype definition because we have $n=0$)

so the left type is a supertype of the right type.

c. supertype:

$$\frac{}{\vdash \{ \} <: \{ \}} \quad \frac{}{\vdash \{ x : \text{Int} \} <: \{ \}}$$

$$\vdash \{ \} \rightarrow \{ x : \text{Int} \} <: \{ \} \rightarrow \{ \}$$

so the left type is a supertype of the right type.

Problem 3.

$r_1 : \{x: \text{Int}, y: \text{Int}\}, r_2 : \{y: \text{Int}, z: \text{Int}\}$

section 1.

see section 2

$n: \text{Int}, r_1: \{x: \text{Int}, y: \text{Int}\}, r_2: \{y: \text{Int}, z: \text{Int}\} \vdash$

$\{x = r_1.x + 1; y = (\text{If } n = 0 \text{ Then } r_1.y - 1 \text{ Else } r_2.y + 1); z = r_2.z + 2\} : \{x: \text{Int}, y: \text{Int}, z: \text{Int}\}$

$n: \text{Int}, r_1: \{x: \text{Int}, y: \text{Int}\} \vdash \text{Function } r_2: \{y: \text{Int}, z: \text{Int}\} \rightarrow$
 $\{x = r_1.x + 1; y = (\text{If } n = 0 \text{ Then } r_1.y - 1 \text{ Else } r_2.y + 1); z = r_2.z + 2\} : \{x: \text{Int}, y: \text{Int}, z: \text{Int}\}$

$n: \text{Int} \vdash \text{Function } r_1: \{x: \text{Int}, y: \text{Int}\} \rightarrow \text{Function } r_2: \{y: \text{Int}, z: \text{Int}\} \rightarrow$
 $\{x = r_1.x + 1; y = (\text{If } n = 0 \text{ Then } r_1.y - 1 \text{ Else } r_2.y + 1); z = r_2.z + 2\} : \{x: \text{Int}, y: \text{Int}, z: \text{Int}\}$

$\emptyset \vdash \text{Function } n: \text{Int} \rightarrow \text{Function } r_1: \{x: \text{Int}, y: \text{Int}\} \rightarrow \text{Function } r_2: \{y: \text{Int}, z: \text{Int}\} \rightarrow$
 $\{x = r_1.x + 1; y = (\text{If } n = 0 \text{ Then } r_1.y - 1 \text{ Else } r_2.y + 1); z = r_2.z + 2\}$

$: \text{Int} \rightarrow \{x: \text{Int}, y: \text{Int}\}$
 $\rightarrow \{y: \text{Int}, z: \text{Int}\}$
 $\rightarrow \{x: \text{Int}, y: \text{Int}, z: \text{Int}\}$

section 2:

now we divide into three evaluations:

We use ... to represent $n: \text{Int}, r_1: \{x: \text{Int}, y: \text{Int}\}, r_2: \{y: \text{Int}, z: \text{Int}\}$

one:

$\frac{\dots \vdash r_1: \{x: \text{Int}, y: \text{Int}\}}{\dots \vdash r_1.x : \text{Int}} \quad \dots \vdash 1 : \text{Int}$
 $\dots \vdash r_1.x + 1 : \text{Int}$

two:

① $\frac{\dots \vdash n : \text{Int} \quad \dots \vdash 0 : \text{Int}}{\dots \vdash n = 0 : \text{Bool}}$

② $\frac{\dots \vdash r_1: \{x: \text{Int}, y: \text{Int}\}}{\dots \vdash r_1.y : \text{Int}} \quad \dots \vdash 1 : \text{Int}$
 $\dots \vdash (r_1.y - 1) : \text{Int}$

③ $\frac{\dots \vdash r_2: \{y: \text{Int}, z: \text{Int}\}}{\dots \vdash r_2.y : \text{Int}} \quad \dots \vdash 1 : \text{Int}$
 $\dots \vdash (r_2.y + 1) : \text{Int}$

from ①, ②, ③, we have:

$\dots \vdash n = 0 : \text{Bool}, \dots \vdash r_1.y - 1 : \text{Int}, \dots \vdash r_2.y + 1 : \text{Int}$
 $\dots \vdash \text{If } n = 0 \text{ Then } (r_1.y - 1) \text{ Else } (r_2.y + 1) : \text{Int}$

three:

$\dots \vdash r_2: \{y: \text{Int}, z: \text{Int}\}$
 $\dots \vdash r_2.z : \text{Int} \quad \dots \vdash 2 : \text{Int}$
 $\dots \vdash r_2.z + 2 : \text{Int}$